



High-Intensity Interval Training Prescribed Within the Secondary Severe-Intensity Domain Improves Critical Speed But Not Finite Distance Capacity

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Abstract

Purpose Improvements in D' (the fatigability constant for running) subsequent to training interventions remain elusive. High-intensity interval training (HIIT) within the severe intensity domain for short durations (< 2-min) have been theorized to improve D' . The purpose of the present study was to assess this in a group of moderately trained individuals.

Methods Eighteen participants completed graded exercise testing (GXT), 40-m sprint testing and a 3-min all-out test (3MT) for running to determine key mechanistic and physiological parameters. Participants were randomly assigned into one of two groups based on intensity prescription ($G_{140\%} = 140\%$ of critical speed [CS]), or time intervals ($G_{90-s} = 90-s$) to complete a twice-weekly training intervention for 6-weeks followed by re-assessment.

Results No between-group differences were present either prior to or following the intervention. Substantial and meaningful improvements were detected during the post-intervention period for both groups for VO_{2max} ($G_{140\%}$: + 7.60%; G_{90-s} : + 11.67%), speed evoking VO_{2max} (sVO_{2max} ; $G_{140\%}$: + 4.33%; G_{90-s} : + 2.92%), gas exchange threshold (GET; $G_{140\%}$: + 12.02%; G_{90-s} : + 20.52%), speed evoking GET ($sGET$; $G_{140\%}$: + 4.17%; G_{90-s} : + 7.92%), CS ($G_{140\%}$: $M = 0.62$ m/s; G_{90-s} : $M = 0.46$ m/s), D' ($G_{140\%}$: $M = -56.34$ m; G_{90-s} : $M = -18.36$ m), FI% ($G_{140\%}$: $M = -6.75\%$; G_{90-s} : $M = -4.38\%$) and maximal distance ($G_{140\%}$: $M = 49.67$ m; G_{90-s} : $M = 58.38$ m).

Conclusions The prescribed intensities and durations were insufficient to elicit improvements in D' . Improvements in D' may be dependent on very short-duration intervals (i.e. < 60 to 90-s) at speeds exceeding 140% CS but below maximal sprint speed.

Keywords Moderately active · Severe intensity · Threshold

Introduction

High-intensity interval training (HIIT) is an effective approach for improving either aerobic or anaerobic metabolism depending on the manipulation of any of the nine interval prescription variables (e.g. work and rest intensity and

duration, exercise modality, number of repetitions, number of series etc.) [4, 5]. The HIIT format may also be used to infer specific anaerobic and/or neuromuscular adaptations when the locomotor speeds typically exceed 95% of the minimal speed eliciting maximal oxygen uptake (sVO_{2max}) through to 100% of maximal sprinting speeds (MSS) [5]. The key starting point for HIIT programming is therefore to define the threshold for “high-intensity” since the metabolic and neuromuscular responses to such intensities, by way of energy contributions from specific bioenergetic pathways, could vary considerably [13, 35].

The American College of Sports Medicine (ACSM), for example, defines “high” intensity as heart rate values between 75 and 96% of the maximum heart rate (HR_{max}) [1]. A valid criticism of such an intensity rating scale is that the lactate response to a given heart rate at the individual level is not accounted for [33, 34]. Other authors however,

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define a “heavy” intensity zone as being between two key physiological thresholds; the gas exchange threshold (GET) at the lower end, and the critical power (CP; for cycling or critical speed [CS] for running) at the higher end; thereby accounting for metabolic perturbations at the said intensity [30, 31]. Although the intensity zone defined by Poole and Jones [30] and Rossiter [31] is arguably more valid and rigorous, a legitimate critique is that it is reliant on substantially more sophisticated equipment and multiple testing bouts that may be beyond the reach of most individuals [25].

A work-around lies in the method of deriving CS. Since CS defines the upper-bound of the heavy-intensity domain, exercising within close proximity to CS would guarantee high-intensity exercise in the absence of knowing the speed evoking the gas exchange threshold (sGET). The traditional method of deriving CS is reliant on multiple testing bouts where the speeds for various distances lasting 2–15 min are plotted and modelled using the classic hyperbolic function; CS being the asymptote, and D' the curvature constant [12, 29]. The D' parameter represents a “fatigability constant”, or more specifically, a finite energy reserve for running at speeds exceeding CS [9, 15]. The latter point is especially pertinent since running at speeds above CS would lead to the speed-dependent depletion of D' , the rate of depletion being reliant on the proximity of running speed to CS, which once wholly expended, would lead to exercise cessation [29]. The tolerable duration of exercise (t_{LIM}) at intensities above CS, termed the “severe intensity domain”, is therefore highly predictable, lending itself well to exercise programming within this intensity domain [17, 30, 37]. Moreover, intensities between 50% Δ (Δ = difference between CS and sVO_{2max}) and 175% CS (termed the secondary severe intensity domain [SSID]) have reported distinct VO_2 responses compared to “lower” intensities, implying that the severe intensity domain itself may be sub-categorized [3]. More importantly, the t_{LIM} of exercise within the severe intensity domain is proportional to the magnitude of D' , therefore surmising that an improvement in D' would extend t_{LIM} and improve performance within this domain. Meaningful improvements in D' , subsequent to HIIT interventions, have however remained elusive (see Table 8.2 from Craig et al. [14]).

In lieu of the substantial methodological variability of the traditional method of deriving CS and D' [25], the 3-min all-out test (3MT) for running may potentially be consigned as the most viable contender for deriving CS and D' [27]. Optimal HIIT prescriptions have been observed when accounting for both CS and D' [7]. Although the utility of the CS concept for interval prescription is well known [17, 28, 37], research using parameters specifically derived from the running 3MT remains severely limited. The dearth of such research is especially pertinent for moderately active individuals. Therefore, the focus of the present study was to

utilize the CS concept to (1) appropriately prescribe HIIT intervals within the SSID and (2) determine whether shorter bouts of fixed duration (e.g. 90-s) or fixed velocity bouts (e.g. 140% CS) would be sufficient to increase the magnitude of D' , and (3) determine the effectiveness of such an approach for modifying, amongst other parameters, the cardiovascular and maximal sprint speed (MSS) responses for moderately active individuals.

Methods

Participants

A sample of moderately active, non-athletic individuals ($N=18$) completed a 6-week HIIT program using the CS concept. Participants were randomly assigned into one of two groups, $G_{140\%}$ ($n=9$; age = 23.05 ± 3.79 years, height = 175.03 ± 8.04 cm, weight = 75.40 ± 12.72 kg), and G_{90-s} ($n=9$; age = 24.48 ± 3.28 years, height = 170.30 ± 11.54 cm, weight = 72.48 ± 13.21 kg), by minimization of differences based on VO_{2max} , CS and D' parameters (see experimental procedures). Written, informed consent to participate in the present study, which was approved by the Institutional Ethics Committee, was provided by all participants. For inclusion into the study, participants had to be classified as at least “moderately” active in accordance with the International Physical Activity Questionnaire (IPAQ) and as “low risk” based on the risk stratification criteria for cardiovascular disease [1]. At the time of testing all participants were healthy and free of injury.

Experimental Design

A pretest–posttest control group study design was utilized [6] whereby participants completed three tests both at baseline and following the 6-week intervention, namely: (1) a graded exercise test (GXT) with a verification bout to determine VO_{2max} , sVO_{2max} , HR_{max} , GET and sGET, (2) a 40-m all-out sprint to primarily determine MSS, 10-m and 40-m split times, and (3) the running 3MT to determine CS and D' as well as other key parameters derived from a bi-exponential model applied to the speed-time response of the 3MT [22]. Each test was separated by 48-h to allow for adequate recovery. Instructions were provided to all participants to: (1) arrive at the laboratory in a well-rested and fully hydrated state, (2) avoid caffeine, alcohol and strenuous exercise for at least 48-h prior to testing, and (3) be at least 3-h post-prandial on the day of testing. Testing was scheduled such that diurnal biological variations could be minimized by keeping testing to within ± 2 h on the separate testing days. Each testing session was preceded by a 10–15 min

warm up led by a qualified strength and conditioning coach to minimize the potential for injury.

Experimental Procedures

Prior to testing, all equipment was calibrated as per manufacturer instructions. Briefly, gas sensors were calibrated with gases of known concentrations (15% O₂, 5% CO₂) which was then verified against ambient air, and the turbine volume transducer was calibrated using a 3 L syringe (Type M 9474-C, Cortex Biophysik, Leipzig, Germany). Pulmonary gas exchange was measured breath-by-breath during the GXT (Metamax 3B, Cortex GmbH, Leipzig, Germany) where participants were fitted with a facemask and breathed through a low-resistance impeller turbine.

The GXT with a verification bout was completed on a motorized treadmill (4Front, Woodway, Waukesha, USA) with the treadmill grade set at 1% to replicate outdoor running conditions [16]. A custom protocol was utilized to determine the appropriate starting speed such that ramp increments of 0.22 m/s/min would achieve exhaustion, and thus VO_{2max}, within approximately 10-min (see Jamnick et al. [23] and Strom et al. [24] for full protocol details). Once exhaustion was reached, a 3-min active recovery period at 1.67 m/s was initiated, followed by a verification bout that commenced at a running speed set at 2-stages below end-stage speed until exhaustion was again achieved [19].

Determination of VO_{2max}

“True” VO_{2max} was recorded as the highest value (using 15-s averaging of the raw data) between the GXT and verification bout if the agreement was within 3%; based on these criteria four participants required re-testing. The GET was determined by: (1) an increase in the ventilatory equivalent of O₂ (V_E/VO₂) without an associated increase in the ventilatory equivalent of CO₂ (V_E/VCO₂), and (2) the first loss in linearity between minute ventilation (V_E) and carbon dioxide production (VCO₂) [2]. Speed evoking GET was determined by interpolation at 1-min preceding GET attainment. Heart rate data were continuously collected via telemetry at 1 Hz (Polar H7, Polar, Kempele, Finland).

40-m All-Out Sprint

A series of three paired photocells (TCI system, Brower Timing Systems, Utah, USA) were set-up to capture the relevant time intervals at the start, 10-m and 40-m marks. Participants started 1-m behind the first set of photocells and were then instructed to run as fast as possible to a set of cones placed 5-m beyond the 40-m mark to encourage full effort over the entire 40-m distance. Displacement data

were continuously recorded and digitized (Tracker, Version 4.11.0, Open Source Physics, Boston, USA) using a high-speed camera that captured video footage at 100 Hz (Sony Cyber-shot DSC-RX10 MK III, Sony, New York, America). The displacement data were numerically differentiated to obtain raw speed-time data which were filtered using a fourth order, zero lag Butterworth filter with a cut-off of 2–6 Hz [38]. The filtered data were then modelled using a mono-exponential function which is applicable from recreational to highly trained sprinters [32] (see Eq. 1):

$$v(t) = MSS \cdot (1 - e^{-t/\tau}) \quad (1)$$

where $v(t)$ is the velocity at any instant in time, MSS is the maximal sprint speed reached at the end of the acceleration, and τ is the velocity time constant. Least squares modelling procedures were used to fit the data from the 40-m sprint test using commercial software (OriginPro 2018b [version 94E], OriginLab, Northampton, USA).

Three-Minute All-Out Exercise Test

All 3MT testing took place on a 400-m outdoor sprinting track at sea-level at an ambient temperature and humidity of 18–22 °C and 40–45% respectively, and wind speeds not exceeding 2 m/s. Prior to testing, participants completed a comprehensive warm-up consisting of light jogging (< sGET), dynamic stretching and progressive speed runs (< 20-m), followed by a 5-min recovery. During the recovery period participants were fitted with a heart rate belt (Polar H7, Kempele, Finland), footpod (Stryd, Boulder, USA) and a wrist-watch with a built-in global positioning system (GPS) (Polar V800, Kempele, Finland) which recorded all heart rate and speed data at 1 Hz.

Participants were instructed to run all-out and to maintain their maximum possible speed throughout the full duration of the test. Strong verbal encouragement was provided throughout, and to minimize the potential for pacing, participants were blinded to the elapsed time or time remaining. The speed-time data obtained from the footpod were exported and modelled in OriginPro using a novel bi-exponential equation (see Eq. 2; Fig. 1) modified from Kramer et al. [22]

$$S(t) = A_g(1 - e^{-t/\tau_g}) - A_d(1 - e^{-t/\tau_d}) \quad (2)$$

where $S(t)$ is the speed at any instant in time, A_g is the exponential growth amplitude, e is a mathematical constant (~ 2.718), τ_g is the time constant of the exponential growth term, A_d is the exponential decay amplitude, and τ_d is the time constant of the exponential decay term. Additional parameters may be derived from Eq. 2 that provide further insights to the 3MT running performance: t_{max} is the time to maximal speed, S_{max} is the magnitude of the highest speed

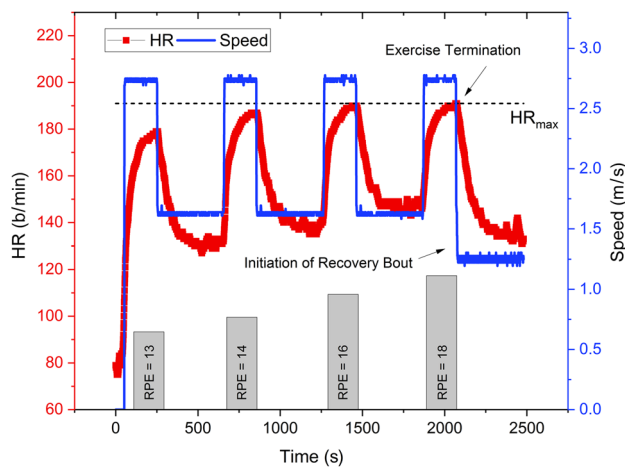


Fig. 1 Heart rate response (red line) to custom HIIT session for a representative participant. Work and recovery speeds (blue line) were calculated from Eqs. 2 or 3 (depending on group). The participants' maximum heart rate is illustrated with a black dotted line is shown for reference. Grey boxes indicate the participants rating of perceived exertion (RPE) for each work interval

attained during the 3MT, CS is the critical speed denoted as the asymptote of the speed decay, D' is the area under the curve above CS , and FI is the fatigue index represented as a ratio (expressed as a percentage) of the differences between S_{max} and CS relative to S_{max} [22]. A non-linear least squares algorithm was used to optimize the model parameters by minimizing the sum of squared errors between the model and the observed data.

HIIT Program Design

High-intensity training sessions were programmed twice weekly, separated by 48-h over a 6-week period. Interval prescriptions differed depending on the group allocation,

although both groups experienced the same fractional D' depletion of 70% per interval, utilizing a 1:2 work-to-rest ratio, and completed recovery intervals at 70% of their CS (see Table 1).

For $G_{140\%}$, intervals were designed with a fixed running speed set a 140% of CS and a variable interval duration (i.e. t_{LIM}) derived using the following equation (modified from [7]):

$$t_{LIM} = 0.7 \cdot D' / (1.4 \cdot CS - CS) = 0.7 \cdot D' / (0.4 \cdot CS) \quad (3)$$

For example, an individual in the $G_{140\%}$ group with a CS of 3.5 m/s and a D' of 160-m would therefore have a prescribed t_{LIM} of 80-s. Stated differently, when running at 4.9 m/s (140% of 3.5 m/s), it would take this individual 80-s to deplete D' by 70%. The recovery interval would last twice as long as the calculated t_{LIM} (1:2 work-to-rest ratio) and was completed at 70% of CS which would allow for some, but not complete, recovery of D' such that the subsequent bout would be initiated in a partially D' -depleted state. Such a protocol enables a total of 4 bouts per session prior to depleting of D' , along with attainment of HR_{max} and subsequent exercise termination [7].

For G_{90-s} , intervals were designed with a pre-set t_{LIM} of 90-s such that the prescribed running speed (i.e. S in Eq. 4) would yield the required 70% D' depletion (see Eq. 4 and Table 1):

$$S = 0.7 \cdot D' / 90s + CS \quad (4)$$

As an example, an individual in the G_{90-s} group with a CS of 3.2 m/s and a D' of 180-m would therefore be prescribed a running interval speed of 4.6 m/s. Stated differently, for this individual, running at 4.6 m/s would ensure that D' would be depleted by 70% by the end of their 90-s interval. The recovery intervals would also be completed with a 1:2 work-to-rest ratio at an intensity of 70% of CS , also ensuring a

Table 1 Work and recovery parameters for both groups

Variable	$G_{140\%}$ ($n=9$)		G_{90-s} ($n=9$)		Interval training group differences ($G_{90-s}-G_{140\%}$)				
	Mean	SD	Mean	SD	M_{diff}	95% CI	P -value	d	95% CI
Int speed (m/s)	4.27	0.72	4.50	0.54	0.23	−0.41, 0.86	0.457	0.34	−0.58, 1.29
Int speed (% CS)	140.01	0.31	146.32	21.78	6.31	−9.08, 21.70	0.397	0.39	−0.53, 1.34
Int speed (% VO_{2max})	115.59	21.42	120.76	10.41	5.17	−11.66, 22.00	0.524	0.29	−0.63, 1.23
Int speed (% $40-mS_{max}$)	58.11	8.63	59.74	4.25	1.63	−5.16, 8.43	0.617	0.23	−0.69, 1.16
Rec speed (m/s)	2.14	0.36	2.18	0.32	0.04	−0.30, 0.38	0.806	0.11	−0.81, 1.04
Rec speed (% CS)	70.01	0.26	69.91	0.26	−0.10	−0.35, 0.16	0.446	−0.35	−1.30, 0.57
Rec speed (% VO_{2max})	57.81	10.77	58.81	9.92	1.01	−9.34, 11.34	0.839	0.09	−0.83, 1.02
Int duration (s)	112.89	53.49	90.00	0.00	−22.89	−64.01, 18.23	0.218	−0.58	−1.54, 0.35
Rec duration (s)	225.89	107.08	180.00	0.00	−45.89	−128.20, 36.42	0.217	−0.58	−1.54, 0.35
Total duration (s)	1355.12	642.30	1080.00	0.00	−275.11	−768.83, 218.61	0.235	—	—

int interval, rec recovery, CS critical speed, VO_{2max} maximal oxygen uptake rate, $40-mS_{max}$ maximal speed attained during the 40-m sprint

total of four bouts per session with full depletion of D' and attainment of HR_{max} in the final bout (see Fig. 1).

Progressive overload was implemented by adjusting the work speed by approximately 5% per week as tolerated. For each session and for each participant the heart rate (Polar H7, Polar, Kempele, Finland), speed (Stryd, Bolder, USA) and rating of perceived exertion (RPE; Borg [6–20]) were continuously monitored and recorded. For safety, we set a speed-limit of 4.44–5.00 m/s depending on the speed tolerance of the individual. To ensure appropriate interval loading subsequent to the imposed speed-limit, adjustments were made to treadmill gradation (see Table 1 in Pettitt et al. [26]).

Statistical Analyses

Between-group differences at baseline were evaluated using independent t-tests. Two-way repeated measures analysis of variance (ANOVA) (group [G_{90-s} and $G_{140\%}$] and time [pre- and post-intervention]) with Tukey post-hoc test was applied for each independent parameter. All data are presented as mean $\pm$ SD, coupled with between-group mean differences ($_bM_{diff}$) and within-group mean differences ($_wM_{diff}$) with the corresponding 95% confidence intervals (95% CI) as well as the unbiased standardized effect size (Cohen's $d_{unbiased}$, with 95% CI) [10]. The α level was set to 0.05 to minimize the type I error rate.

All analyses were conducted using Statistica version 11 (StatSoft, Dell Software, California, USA), OriginPro version 2018b (OriginLab, Northampton, USA) and ESCI version 2016 (Free Software Foundation, Melbourne, AUS).

Results

Participants and Training Program

Group allocations were hinged on minimization of between-group differences at baseline for VO_{2max} , CS and D' (see Table 2). No meaningful between-group differences were observed for any of the laboratory-based or field-based parameters at baseline, with the exception of body mass in favor of G_{90-s} ($M_{G_{90-s}} - M_{G_{140\%}} = 13.37$ kg, 95% CI [3.52, 22.22]).

Despite the differences in the design of the intervention program (i.e. speed-based [$G_{140\%}$] vs. time-based [G_{90-s}]), there were no meaningful between-group differences for work or recovery speeds, nor work or recovery durations (see Table 2). Note should however be taken of the variability and between-group mean difference (M_{diff})

for parameters such as the prescribed interval speed as a function of %GET and % VO_{2max} , as well as the work and recovery interval durations. Standardising the interval durations (i.e. G_{90-s} method) may be more conducive for larger groups and/or team-based sports.

Graded Exercise Test

Following the intervention, substantial improvements were noted for laboratory-based parameters such as VO_{2max} , sVO_{2max} , GET and $sGET$ for $G_{140\%}$ (+7.60%, +4.33%, +12.02%, and +4.17%) and G_{90-s} (+11.67%, +2.92%, +20.52%, and +7.92%) respectively (see Table 2). The 95% CI of the standardized effect sizes (SES; denoted by $d_{unbiased}$) were fairly long for each of these parameters which ranged from small to large. Interestingly, a meaningful improvement was noted for %GET for G_{90-s} , showing a ~5% increase following the intervention which was not noted for $G_{140\%}$. The SES for this improvement however was small ($d_{unbiased} = 0.47$, 95% CI [0.07, 0.99]) with a wide confidence interval.

Field Tests

Following the intervention, the 40-m sprint-specific parameters showed no meaningful improvements in MSS or 40-m time. An exception here is the 10-m split time for G_{90-s} which showed a 2.5% performance decrement following the intervention. The SES for this improvement was however small with a moderately long confidence interval ($d_{unbiased} = 0.24$, 95% CI [0.05, 0.46]). The overall implication here is that sprint performances remained intact subsequent to the intervention.

Parameters related to the 3MT however showed substantial deviations from baseline after the intervention, with changes being independent of group allocation (see Table 2). For both groups the most notable improvements were recorded for CS ($G_{140\%}$: $M = 0.62$ m/s, 95% CI [0.40, 0.84]; G_{90-s} : $M = 0.46$ m/s, 95% CI [0.24, 0.68]), FI ($G_{140\%}$: $M = -6.75\%$, 95% CI [-10.45, -3.05]; G_{90-s} : $M = -4.38\%$, 95% CI [-8.08, -0.68]), maximal running distance ($G_{140\%}$: $M = 49.67$ m, 95% CI [26.71, 72.63]; G_{90-s} : $M = 58.38$ m, 95% CI [35.42, 72.63]), distance covered in the second minute ($G_{140\%}$: $M = 18.20$ m, 95% CI [6.45, 29.95]; G_{90-s} : $M = 29.10$ m, 95% CI [17.36, 40.85]), and distance covered in the final minute ($G_{140\%}$: $M = 22.28$ m, 95% CI [7.76, 36.80]; G_{90-s} : $M = 26.30$ m, 95% CI [11.78, 40.82]). A substantial performance decrement was noted for D' , specifically for participants from $G_{140\%}$ ($G_{140\%}$: $M = -56.34$ m, 95% CI [-88.57, -24.11]; G_{90-s} : $M = -18.36$ m, 95% CI [-50.59, 13.87]). Interestingly, for $G_{140\%}$ the τ_d parameter showed a substantial decrease ($M = -11.04$ s, 95% CI [-18.46, -3.61]) implying faster speed decay towards CS, whereas

Table 2 Repeated measures ANOVA results for between-group ($G_{140\%}$ vs. G_{90-s}) and within-group (pre- vs. post-intervention) differences

Variable	Group	Pre	Post	M_{diff} (M, 95% CI)	Tukey P -value	$d_{unbiased}$ (95% CI)	F (I_{Vdf} , $errdf$, P)
Laboratory-based data (anthropometry and graded exercise test)							
Height (cm)	$G_{140\%}$ (M, SD)	162.90 (6.97)	162.84 (7.29)	−0.06 (−0.67, 0.56)	0.851	0.00 (−0.10, 0.08)	Time
	G_{90-s} (M, SD)	170.22 (10.04)	170.46 (9.71)	0.23 (−0.38, 0.85)	0.433	0.02 (−0.03, 0.07)	$F(1,16)=0.19$, $P=0.671$
	M_{diff} (M, 95% CI)	−7.32 (−15.93, 1.29)	−7.61 (−16.22, 1.00)				Time × Group
	Tukey P -value	0.090	0.079				$F(1,16)=0.50$, $P=0.492$
	$d_{unbiased}$ (95% CI)	0.81 (−0.13, 1.80)	0.84 (−0.10, 1.85)				
Weight (kg)	$G_{140\%}$ (M, SD)	64.91 (7.48)	64.68 (7.79)	−0.23 (−1.13, 0.67)	0.590	−0.03 (−0.13, 0.07)	Time
	G_{90-s} (M, SD)	78.28 (11.76)	77.42 (11.48)	−0.86 (−1.76, 0.04)	0.061	−0.07 (−0.16, 0.01)	$F(1,16)=3.29$, $P=0.088$
	M_{diff} (M, 95% CI)	−13.37 (−23.19, −3.54)	−12.74 (−22.57, −2.92)				Time × Group
	Tukey P -value	0.011	0.014				$F(1,16)=1.08$, $P=0.315$
	$d_{unbiased}$ (95% CI)	1.29 (0.31, 2.37)	1.24 (0.26, 2.31)				
HR_{max} (b/min)	$G_{140\%}$ (M, SD)	187.56 (6.33)	188.67 (10.30)	1.11 (−3.82, 6.04)	0.639	0.12 (−0.55, 0.81)	Time
	G_{90-s} (M, SD)	192.44 (9.07)	191.67 (7.23)	−0.78 (−5.71, 4.15)	0.742	−0.09 (−0.44, 0.26)	$F(1,16)=0.01$, $P=0.921$
	M_{diff} (M, 95% CI)	−4.89 (−13.26, 3.48)	−3.00 (−11.37, 5.37)				Time × Group
	Tukey P -value	0.233	0.458				$F(1,16)=0.33$, $P=0.574$
	$d_{unbiased}$ (95% CI)	0.60 (−0.33, 1.57)	0.32 (−0.60, 1.26)				
VO_{2max} (mL/kg/min)	$G_{140\%}$ (M, SD)	46.24 (6.64)	50.04 (10.05)	3.81 (1.36, 6.25)	0.005	0.40 (0.05, 0.83)	Time
	G_{90-s} (M, SD)	44.24 (5.44)	49.44 (6.71)	5.19 (2.75, 7.64)	<0.001	0.77 (0.36, 1.32)	$F(1,16)=30.39$, $P<0.001$
	M_{diff} (M, 95% CI)	1.99 (−5.42, 9.40)	0.61 (−6.80, 8.02)				Time × Group
	Tukey P -value	0.576	0.864				$F(1,16)=0.72$, $P=0.409$
	$d_{unbiased}$ (95% CI)	−0.31 (−1.25, 0.61)	−0.07 (−0.99, 0.85)				
sVO_{2max} (m/s)	$G_{140\%}$ (M, SD)	3.79 (0.94)	3.93 (0.86)	0.14 (0.01, 0.26)	0.031	0.38 (0.09, 0.73)	Time
	G_{90-s} (M, SD)	3.77 (0.65)	3.89 (0.75)	0.12 (−0.01, 0.24)	<0.001	0.70 (0.29, 1.24)	$F(1,16)=9.54$, $P=0.007$
	M_{diff} (M, 95% CI)	0.02 (−0.78, 0.83)	0.05 (−0.76, 0.85)				Time × Group
	Tukey P -value	0.949	0.904				$F(1,16)=0.07$, $P=0.798$
	$d_{unbiased}$ (95% CI)	−0.03 (−0.95, 0.89)	−0.05 (−0.98, 0.87)				

Table 2 (continued)

Variable	Group	Pre	Post	wM_{diff} (M, 95% CI)	Tukey <i>P</i> -value	$d_{unbiased}$ (95% CI)	$F(I_{Vdf}, errdf, P)$
GET (mL/kg/min)	G _{140%} (M, SD)	29.87 (7.96)	33.95 (11.26)	4.08 (1.54, 6.63)	0.004	0.38 (0.09, 0.73)	Time
	G _{90-s} (M, SD)	30.44 (7.11)	36.60 (8.73)	6.17 (3.62, 8.71)	<0.001	0.70 (0.29, 1.24)	$F(1,16) = 36.4$, $P < 0.001$
	bM_{diff} (M, 95% CI)	−0.57 (−9.47, 8.32)	−2.65 (−11.55, 6.24)				Time × Group
	Tukey <i>P</i> -value	0.893	0.536				$F(1,16) = 1.50$, $P = 0.238$
	$d_{unbiased}$ (95% CI)	0.07 (−0.85, 0.99)	0.25 (−0.67, 1.19)				
GET as %VO ₂ (%)	G _{140%} (M, SD)	63.77 (8.28)	66.35 (8.57)	2.59 (−0.93, 6.10)	0.138	0.28 (−0.01, 0.61)	Time
	G _{90-s} (M, SD)	68.08 (9.83)	73.10 (9.39)	5.02 (1.50, 8.53)	0.008	0.47 (0.07, 0.99)	$F(1,16) = 10.53$, $P = 0.005$
	bM_{diff} (M, 95% CI)	−4.31 (−13.35, 4.72)	−6.74 (−15.78, 2.29)				Time × Group
	Tukey <i>P</i> -value	0.326	0.133				$F(1,16) = 1.08$, $P = 0.315$
	$d_{unbiased}$ (95% CI)	0.45 (−0.47, 1.41)	0.71 (−0.22, 1.70)				
sGET (m/s)	G _{140%} (M, SD)	2.48 (0.92)	2.67 (0.93)	0.20 (0.05, 0.34)	0.011	−0.04 (−0.01, 0.03)	Time
	G _{90-s} (M, SD)	2.60 (0.70)	2.89 (0.83)	0.29 (0.15, 0.44)	<0.001	0.02 (−0.03, 0.07)	$F(1,16) = 25.34$, $P < 0.001$
	bM_{diff} (M, 95% CI)	−0.12 (−0.97, 0.73)	−0.22 (−1.07, 0.63)				Time × Group
	Tukey <i>P</i> -value	0.764	0.594				$F(1,16) = 0.97$, $P = 0.340$
	$d_{unbiased}$ (95% CI)	0.14 (−0.78, 1.07)	0.24 (−0.69, 1.17)				
Field-based data (40-m sprint)							
MSS (m/s)	G _{140%} (M, SD)	7.47 (1.36)	7.43 (1.37)	−0.04 (−0.23, 0.15)	0.659	−0.03 (−0.12, 0.06)	Time
	G _{90-s} (M, SD)	7.58 (1.12)	7.58 (1.11)	−0.01 (−0.19, 0.18)	0.927	−0.01 (−0.20, 0.18)	$F(1,16) = 0.15$, $P = 0.706$
	bM_{diff} (M, 95% CI)	−0.12 (−1.36, 1.13)	−0.15 (−1.39, 1.10)				Time × Group
	Tukey <i>P</i> -value	0.844	0.804				$F(1,16) = 0.06$, $P = 0.804$
	$d_{unbiased}$ (95% CI)	0.09 (−0.83, 1.02)	0.11 (−0.81, 1.04)				
10-m (s)	G _{140%} (M, SD)	1.89 (0.21)	1.88 (0.16)	−0.01 (−0.06, 0.04)	0.717	−0.05 (−0.38, 0.28)	Time
	G _{90-s} (M, SD)	1.89 (0.19)	1.95 (0.22)	0.05 (0.00, 0.11)	0.054	0.24 (0.05, 0.46)	$F(1,16) = 1.46$, $P = 0.244$
	bM_{diff} (M, 95% CI)	0.00 (−0.2, 0.19)	−0.06 (−0.26, 0.13)				Time × Group
	Tukey <i>P</i> -value	0.993	0.504				$F(1,16) = 3.00$, $P = 0.103$
	$d_{unbiased}$ (95% CI)	0.00 (−0.92, 0.93)	0.31 (−0.61, 1.26)				

Table 2 (continued)

Variable	Group	Pre	Post	wM_{diff} (M, 95% CI)	Tukey <i>P</i> -value	$d_{unbiased}$ (95% CI)	$F(I_{Vdf}, errdf, P)$
40-m (s)	G _{140%} (M, SD)	6.17 (0.92)	6.15 (0.84)	−0.02 (−0.17, 0.13)	0.796	−0.02 (0.19, 0.14)	Time
	G _{90-s} (M, SD)	6.05 (0.86)	6.11 (0.84)	0.06 (−0.09, 0.21)	0.443	0.06 (−0.10, 0.23)	$F(1,16)=0.14$, $P=0.716$
	bM_{diff} (M, 95% CI)	0.12 (−0.74, 0.98)	0.04 (−0.82, 0.91)				Time × Group
	Tukey <i>P</i> -value	0.774	0.915				$F(1,16)=0.55$, $P=0.468$
	$d_{unbiased}$ (95% CI)	−0.13 (−1.06, 0.79)	−0.05 (−0.98, 0.87)				
Field-based data (3MT)							
CS (m/s)	G _{140%} (M, SD)	3.05 (0.52)	3.67 (0.57)	0.62 (0.40, 0.84)	<0.001	1.03 (0.46, 1.80)	Time
	G _{90-s} (M, SD)	3.11 (0.46)	3.57 (0.57)	0.46 (0.24, 0.68)	<0.001	0.81 (0.30, 1.46)	$F(1,16)=54.56$, $P<0.001$
	bM_{diff} (M, 95% CI)	−0.06 (−0.59, 0.47)	0.10 (−0.43, 0.63)				Time × Group
	Tukey <i>P</i> -value	0.809	0.701				$F(1,16)=1.18$, $P=0.294$
	$d_{unbiased}$ (95% CI)	0.12 (−0.80, 1.05)	−0.16 (−1.10, 0.76)				
D' (m)	G _{140%} (M, SD)	188.88 (74.76)	132.54 (44.29)	−56.34 (−88.57, −24.11)	0.002	−0.83 (−1.58, −0.22)	Time
	G _{90-s} (M, SD)	175.47 (53.60)	157.11 (43.97)	−18.36 (−50.59, 13.87)	0.254	−0.34 (−0.96, 0.23)	$F(1,16)=12.07$, $P=0.003$
	bM_{diff} (M, 95% CI)	13.41 (−42.14, 68.95)	−24.57 (−80.12, 30.97)				Time × Group
	Tukey <i>P</i> -value	0.616	0.362				$F(1,16)=3.12$, $P=0.096$
	$d_{unbiased}$ (95% CI)	−0.20 (−1.13, 0.72)	0.53 (−0.40, 1.49)				
FI (%)	G _{140%} (M, SD)	56.00 (7.35)	49.25 (5.12)	−6.75 (−10.45, −3.05)	0.001	−0.96 (−1.85, −0.24)	Time
	G _{90-s} (M, SD)	56.42 (6.84)	52.04 (6.98)	−4.38 (−8.08, −0.68)	0.023	−0.57 (−1.13, −0.12)	$F(1,16)=20.33$, $P<0.001$
	bM_{diff} (M, 95% CI)	−0.42 (−7.04, 6.21)	−2.79 (−9.42, 3.83)				Time × Group
	Tukey <i>P</i> -value	0.895	0.385				$F(1,16)=0.92$, $P=0.351$
	$d_{unbiased}$ (95% CI)	0.06 (−0.87, 0.92)	0.43 (−0.49, 1.39)				
D _{max} (m)	G _{140%} (M, SD)	730.03 (101.51)	779.7 (130.83)	49.67 (26.71, 72.63)	<0.001	0.38 (0.12, 0.71)	Time
	G _{90-s} (M, SD)	738.11 (85.61)	796.49 (84.38)	58.38 (35.42, 81.34)	<0.001	0.62 (0.30, 1.07)	$F(1,16)=49.77$, $P<0.001$
	bM_{diff} (M, 95% CI)	−8.08 (−110.32, 94.17)	−16.79 (−119.04, 85.46)				Time × Group
	Tukey <i>P</i> -value	0.869	0.732				$F(1,16)=0.32$, $P=0.577$
	$d_{unbiased}$ (95% CI)	0.08 (−0.84, 1.01)	0.15 (−0.78, 1.08)				

Table 2 (continued)

Variable	Group	Pre	Post	wM_{diff} (M, 95% CI)	Tukey <i>P</i> -value	$d_{unbiased}$ (95% CI)	$F(I_V df, err df, P)$
D_{0-60} (m)	$G_{140\%}$ (M, SD)	318.12 (52.59)	327.31 (61.14)	9.19 (− 7.24, 25.62)	0.253	0.15 (− 0.07, 0.38)	Time
	G_{90-s} (M, SD)	327.11 (38.48)	330.09 (30.15)	2.98 (− 13.46, 19.41)	0.706	0.08 (− 0.43, 0.60)	$F(1,16)=1.23$, $P=0.284$
	bM_{diff} (M, 95% CI)	− 8.99 (− 56.11, 38.13)	− 2.78 (− 49.9, 44.35)				Time × Group
	Tukey <i>P</i> -value	0.691	0.902				$F(1,16)=0.32$, $P=0.579$
	$d_{unbiased}$ (95% CI)	0.19 (− 0.73, 1.12)	0.06 (− 0.87, 0.98)				
D_{60-120} (m)	$G_{140\%}$ (M, SD)	216.88 (29.68)	235.08 (34.36)	18.20 (6.45, 29.95)	0.005	0.51 (0.10, 1.01)	Time
	G_{90-s} (M, SD)	215.11 (28.32)	244.21 (28.5)	29.10 (17.36, 40.85)	< 0.001	0.93 (0.41, 1.62)	$F(1,16)=36.44$, $P<0.001$
	bM_{diff} (M, 95% CI)	1.77 (− 28.53, 32.06)	− 9.13 (− 39.43, 21.16)				Time × Group
	Tukey <i>P</i> -value	0.903	0.532				$F(1,16)=1.94$, $P=0.183$
	$d_{unbiased}$ (95% CI)	− 0.06 (− 0.98, 0.86)	0.28 (− 0.64, 1.21)				
$D_{120-180}$ (m)	$G_{140\%}$ (M, SD)	195.03 (27.81)	217.31 (38.87)	22.28 (7.76, 36.80)	0.005	0.60 (0.13, 1.16)	Time
	G_{90-s} (M, SD)	195.89 (28)	222.19 (39.22)	26.30 (11.78, 40.82)	0.001	0.70 (0.23, 1.29)	$F(1,16)=25.16$, $P<0.001$
	bM_{diff} (M, 95% CI)	− 0.86 (− 34.77, 33.06)	− 4.88 (− 38.79, 29.04)				Time × Group
	Tukey <i>P</i> -value	0.958	0.764				$F(1,16)=0.17$, $P=0.683$
	$d_{unbiased}$ (95% CI)	0.03 (− 0.89, 0.95)	0.12 (− 0.80, 1.05)				
HR_{max} (b/min)	$G_{140\%}$ (M, SD)	177.56 (11.06)	177.67 (8.76)	0.11 (− 14.67, 14.90)	0.987	0.29 (− 0.54, 1.15)	Time
	G_{90-s} (M, SD)	186.11 (9.13)	176.33 (27.80)	− 9.78 (− 24.56, 5.01)	0.180	− 0.17 (− 0.49, 0.14)	$F(1,16)=0.96$, $P=0.342$
	bM_{diff} (M, 95% CI)	− 8.56 (− 24.79, 7.68)	1.33 (− 14.9, 17.57)				Time × Group
	Tukey <i>P</i> -value	0.280	0.864				$F(1,16)=1.01$, $P=0.331$
	$d_{unbiased}$ (95% CI)	0.80 (− 0.14, 1.80)	− 0.06 (− 0.99, 0.86)				
A_g (m/s)	$G_{140\%}$ (M, SD)	8.09 (1.35)	8.52 (1.67)	0.42 (− 0.33, 1.17)	0.248	0.25 (− 0.19, 0.73)	Time
	G_{90-s} (M, SD)	8.56 (0.79)	8.92 (1.43)	0.37 (− 0.38, 1.12)	0.314	0.29 (− 0.34, 0.96)	$F(1,16)=2.5$, $P=0.133$
	bM_{diff} (M, 95% CI)	− 0.46 (− 1.81, 0.88)	− 0.41 (− 1.75, 0.94)				Time × Group
	Tukey <i>P</i> -value	0.477	0.532				$F(1,16)=0.01$, $P=0.912$
	$d_{unbiased}$ (95% CI)	0.40 (− 0.52, 1.35)	0.25 (− 0.67, 1.19)				
A_d (m/s)	$G_{140\%}$ (M, SD)	5.04 (1.04)	4.91 (1.17)	− 0.14 (− 0.88, 0.60)	0.699	− 0.11 (− 0.68, 0.44)	Time

Table 2 (continued)

Variable	Group	Pre	Post	${}_wM_{diff}$ (M, 95% CI)	Tukey <i>P</i> -value	$d_{unbiased}$ (95% CI)	$F(I_{Vdf}, errdf, P)$
τ_g (s)	G _{90-s} (M, SD)	5.45 (0.91)	5.35 (1.39)	−0.10 (−0.84, 0.64)	0.774	−0.08 (−0.72, 0.56)	$F(1,16)=0.24$, $P=0.634$
	${}_bM_{diff}$ (M, 95% CI)	−0.41 (−1.55, 0.73)	−0.45 (−1.59, 0.69)				Time × Group
	Tukey <i>P</i> -value	0.456	0.419				$F(1,16)=0.01$, $P=0.944$
	$d_{unbiased}$ (95% CI)	0.40 (−0.52, 1.35)	0.33 (−0.59, 1.27)				
	G _{140%} (M, SD)	1.99 (0.86)	1.96 (0.87)	−0.03 (−1.09, 1.04)	0.956	−0.03 (−0.87, 0.81)	Time
	G _{90-s} (M, SD)	2.26 (1.58)	2.08 (1.19)	−0.18 (−1.24, 0.89)	0.731	−0.11 (−0.98, 0.73)	$F(1,16)=0.08$, $P=0.777$
	${}_bM_{diff}$ (M, 95% CI)	−0.27 (−1.43, 0.89)	−0.12 (−1.28, 1.04)				Time × Group
	Tukey <i>P</i> -value	0.633	0.832				$F(1,16)=0.04$, $P=0.837$
τ_d (s)	$d_{unbiased}$ (95% CI)	0.20 (−0.72, 1.13)	0.11 (−0.81, 1.04)				
	G _{140%} (M, SD)	41.70 (15.38)	30.67 (8.55)	−11.04 (−18.46, −3.61)	0.006	−0.80 (−1.64, −0.10)	Time
	G _{90-s} (M, SD)	35.89 (6.66)	34.01 (8.13)	−1.88 (−9.30, 5.55)	0.599	−0.23 (−1.01, 0.52)	$F(1,16)=6.8$, $P=0.019$
	${}_bM_{diff}$ (M, 95% CI)	5.82 (−4.43, 16.06)	−3.34 (−13.58, 6.9)				Time × Group
	Tukey <i>P</i> -value	0.246	0.499				$F(1,16)=3.42$, $P=0.083$
	$d_{unbiased}$ (95% CI)	−0.47 (−1.42, 0.46)	0.38 (−0.54, 1.33)				
	G _{140%} (M, SD)	7.09 (2.25)	6.78 (2.21)	−0.31 (−2.8, 2.17)	0.793	−0.13 (−0.99, 0.72)	Time
	G _{90-s} (M, SD)	7.30 (3.47)	6.92 (2.57)	−0.38 (−2.86, 2.11)	0.751	−0.11 (−0.99, 0.75)	$F(1,16)=0.17$, $P=0.683$
t_{max} (s)	${}_bM_{diff}$ (M, 95% CI)	−0.21 (−2.88, 2.46)	−0.14 (−2.81, 2.53)				Time × Group
	Tukey <i>P</i> -value	0.871	0.911				$F(1,16)=0.00$, $P=0.969$
	$d_{unbiased}$ (95% CI)	0.07 (−0.85, 0.99)	0.06 (−0.87, 0.98)				
	G _{140%} (M, SD)	7.01 (1.23)	7.33 (1.56)	0.32 (−0.01, 0.64)	0.055	0.20 (−0.09, 0.53)	Time
	G _{90-s} (M, SD)	7.24 (1.04)	7.51 (0.99)	0.27 (0.06, 0.59)	0.100	0.24 (0.09, 0.44)	$F(1,16)=7.26$, $P=0.016$
	${}_bM_{diff}$ (M, 95% CI)	−0.23 (−1.45, 1.00)	−0.18 (−1.40, 1.05)				Time × Group
	Tukey <i>P</i> -value	0.701	0.764				$F(1,16)=0.05$, $P=0.822$
	$d_{unbiased}$ (95% CI)	0.19 (−0.73, 1.12)	0.13 (−0.79, 1.06)				
S_{max} (m/s)	G _{140%} (M, SD)	7.01 (1.23)	7.33 (1.56)	0.32 (−0.01, 0.64)	0.055	0.20 (−0.09, 0.53)	Time
	G _{90-s} (M, SD)	7.24 (1.04)	7.51 (0.99)	0.27 (0.06, 0.59)	0.100	0.24 (0.09, 0.44)	$F(1,16)=7.26$, $P=0.016$
	${}_bM_{diff}$ (M, 95% CI)	−0.23 (−1.45, 1.00)	−0.18 (−1.40, 1.05)				Time × Group
	Tukey <i>P</i> -value	0.701	0.764				$F(1,16)=0.05$, $P=0.822$
	$d_{unbiased}$ (95% CI)	0.19 (−0.73, 1.12)	0.13 (−0.79, 1.06)				
	G _{140%} (M, SD)	7.01 (1.23)	7.33 (1.56)	0.32 (−0.01, 0.64)	0.055	0.20 (−0.09, 0.53)	Time
	G _{90-s} (M, SD)	7.24 (1.04)	7.51 (0.99)	0.27 (0.06, 0.59)	0.100	0.24 (0.09, 0.44)	$F(1,16)=7.26$, $P=0.016$
	${}_bM_{diff}$ (M, 95% CI)	−0.23 (−1.45, 1.00)	−0.18 (−1.40, 1.05)				Time × Group

VO_{2max} maximal oxygen uptake, sVO_{2max} speed evoking VO_{2max} , GET gas exchange threshold, $sGET$ speed evoking GET , CS critical speed, D' fatigability constant, FI fatigue index, D_{max} maximum distance achieved, D_{0-60} distance achieved in first minute, D_{60-120} distance achieved in second minute, $D_{120-180}$ distance achieved in third minute, MSS maximal sprint speed, HR_{max} maximal heart rate, A_g growth amplitude, A_d decay amplitude, τ_g growth time constant, τ_d decay time constant, t_{max} time at which S_{max} was achieved, S_{max} maximal speed attained during the 3MT, ${}_bM_{diff}$ mean between-group difference if read vertically, ${}_wM_{diff}$ mean within-group difference if read horizontally, $d_{unbiased}$ unbiased form of Cohens d for $n < 50$, I_{Vdf} independent variable degrees of freedom, $errdf$ error degrees of freedom

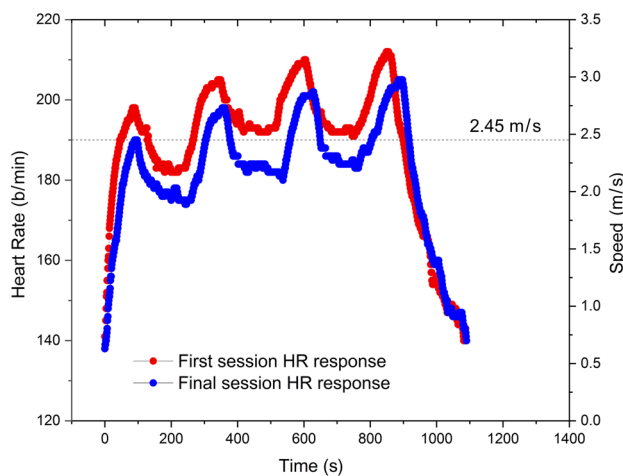


Fig. 2 Differences in the heart rate response for a representative participant. The average interval speed for the first session and final session (black dotted line) is shown, indicating the adaptive response to the same exercise load after a 6-week period

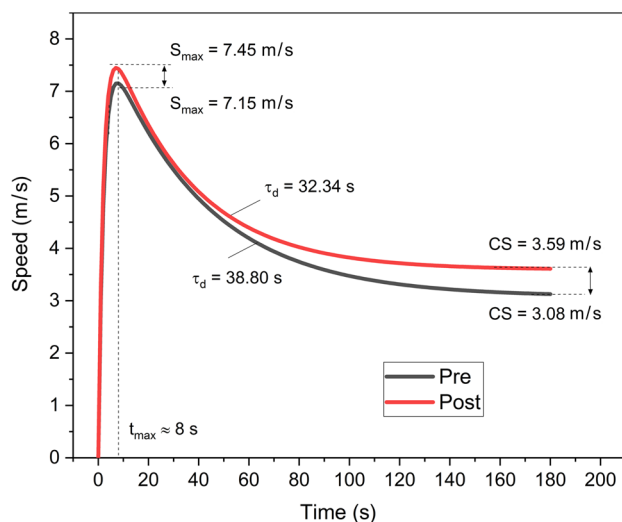


Fig. 3 Post-Pre differences in 3MT performance for all participants. Comparative variables derived from the bi-exponential model are shown for contrast. Note: participant values were pooled due to lack of between-group differences

G_{90-s} exhibited a meaningfully higher S_{max} ($M = 0.27$ m/s, 95% CI [0.06, 0.59]) suggesting greater maximal speed transfer following post-intervention testing.

Work and recovery parameters for the prescribed training intervals are shown in Table 1. The fractional depletion of D' used for the present study ensured that participants reached t_{LIM} within four bouts upon which HR_{max} was reached at the end of the 4th bout. Only two participants were able to achieve a 5th bout, which may indicate an under-estimated CS from the 3MT or enhanced D' recovery kinetics.

Representative data related to the interval heart rate responses are shown in Fig. 2. The first and final sessions are

compared to determine whether, on average, the heart rate response differed. At the same relative intensity, a clear shift in the heart rate response can be seen. The implication here is that for the same relative intensity, the heart rate response was lower thereby indicating the presence of cardiovascular adaptation following the intervention.

The post-pre intervention 3MT speed-time curve data for the group as a whole (since there were no significant between-group differences) is shown in Fig. 3. Clear changes in the speed-time profile, as revealed from the bi-exponential model, are evident with specific emphasis on the upwards shift in CS and maximal distance, coupled with the commensurate decrease in both D' and FI%.

Discussion

The present study demonstrates that exercise training within the secondary severe intensity domain (SSID) yielded improvements in VO_{2max} , sVO_{2max} , $sGET$, CS and maximal distances achieved during the 3MT, whereas substantial decrements were observed for D' and FI%.

The initial postulate was that shorter duration, high-intensity training, as prescribed for the present study, would be sufficient to induce changes in D' . Although the inverse was true, four of 18 participants experienced improvements in D' ranging from 12 to 33%. Three of these participants were from G_{90-s} (using 90-s intervals), and one was from $G_{140\%}$. Interestingly, the only participant from $G_{140\%}$ that showed improvements in D' had the shortest work-interval bouts (i.e. 44-s intervals). Such a finding may lend credence to the postulate that even shorter intervals (i.e. < 60 to 90-s) might be required to elicit favorable changes in D' , although such intensities would mandate a transition from HIIT to sprint-interval training (SIT) [5]. It is worth noting that, based on the methods used to calculate the required interval speed with the commensurate fractional D' depletion, the prescribed running speeds may exceed the physiological capability of moderately trained individuals.

Presently, no field test exists to objectively quantify GET, a key physiological threshold that separates moderate from heavy intensity exercise on the basis of the unique metabolic responses below or above the GET threshold [30]. However, the 3MT is a field test with the means of objectively and reliably deriving CS, an arguably more important physiological threshold, that separates sustainable from non-sustainable exercise (i.e. heavy from severe intensity exercise) [9, 29]. The utility of such a test is unequivocal as it may be used not only as an assessment tool [23], but also as a tool for performance prescription [7, 8] and performance classification [21]; attributes not inherently present in most other field tests.

Much is known about the physiological underpinnings of CS, to the extent that it is conclusively classified as a

“critical metabolic rate” rather than simply a measure of external workload [9, 29]. The same cannot be said for D' , which is categorized as a “fatigability constant” representative of a finite, depletable energy reserve that is predictably expended when speeds exceed CS [9]. The latter fact is crucial for exercise programming on the basis that training intervals can be reliably and predictably prescribed, as was demonstrated in the present study. In other words, for ~90% of individuals in the present study, the participant-specific t_{LIM} and HR_{max} were achieved towards the end of the 4th bout, irrespective of group allocation. Such consistency is remarkable given (1) the varied CS and D' parameters, (2) the ease with which such training intervals are determined for each participant, and (3) our limited understanding of D' on- and off-transient kinetics. The achievement of HR_{max} can, in part, be explained by the putative metabolic perturbations (e.g. elevated intramuscular inorganic phosphate [P_i] and [H^+]) that accumulate during exercise within the severe intensity domain and continue to drive both VO_2 and HR towards maximum values [30, 35].

Originally, D' was thought to represent an anaerobic energy reserve, which although appealing, is now known to be incorrect [11, 29]. Researchers have tried to understand the mechanistic bases of D' with various degrees of success (see Table 8, Craig et al. [14]), but none have yet managed to ascribe D' to any particular physiological parameter(s) with unequivocal certainty. Appropriate exercise prescription that emphatically increases D' to an appreciable, predictable extent currently remains elusive. The present study contributes to the programming-related understanding of D' by demonstrating that exercise within the SSID is insufficient to positively manipulate D' ; the current protocol actually appreciably decreases D' . Alternatively, the present data indicate that exercise within the SSID evokes improvements in aerobic parameters without compromising traditional anaerobic parameters such as MSS, S_{max} (during the 3MT) or 40-m sprint times. Moreover, the improvements noted in the present study were achieved with twice weekly exercise sessions lasting approximately 18–22 min per session depending on group allocation. It is important to note however, that the total session durations to achieve such improvements varied more considerably for $G_{140\%}$ ($M = 22.59$ min, $SD = 10.71$) compared to G_{90-s} ($M = 18$ min, $SD = 0$) (see Table 1). Improvements in VO_{2max} (i.e. +13%), GET (i.e. +21%) and left ventricular thickness (+18%) have been achieved with as few as 1-session per week over a 3-month training period utilizing the HIIT methodology, thereby showing the intensity-dependent nature of VO_{2max} [26]. High-exercise intensities tax both the neuromuscular and metabolic systems, which if appropriately controlled for, would lead to favorable adaptations in both aerobic and anaerobic bioenergetic pathways [26]; both of which were observed in the present study, although clearly skewed more

towards aerobic adaptations. Optimal neuromuscular loading conducive to performance outcomes (e.g. high-speed running tolerance) therefore appears to occur at intensities between 85 and 120% sVO_{2max} [5]. Such findings are reflected in mean interval speeds utilized the present study which occurred within the upper range of this limit (see Table 1), potentially explaining the enhanced aerobic adaptation coupled with the retention of high-speed running capacity.

Unfortunately, the optimum protocol for evoking training improvements in D' still remain elusive. The link between D' and thigh circumference [20], as well as the considerably larger D' values discovered in high-level male rugby players [21], provide some enticing insights related to the potential mechanisms by which D' may be enhanced. Notwithstanding, the fact that the cycling equivalent of D' (termed W') may be potentiated with creatine supplementation [24, 36], and that the W' kinetics may be modelled on the basis of PCr kinetics [18], hints at a peripheral fatigue connection that seems to be associated with the VO_2 slow component that develops when intensities exceed CS/CP [29]. Clearly, more research related to the 3MT methodology, with specific emphasis on D' , is needed.

Conclusions

Improvements in D' may be dependent on very short-duration intervals (i.e. < 60 to 90-s) at speeds exceeding 140% CS but below MSS. Fractional depletion of D' (e.g. 70%) results in predictable achievement of t_{LIM} and HR_{max} within four bouts. Standardizing bouts by using either time-based or speed-based prescriptions yield similar results across both laboratory and field test data. The choice of prescription may therefore depend on the requirements of the coach and/or athlete. Substantial improvements in aerobic parameters may be achieved within a 6-week period without compromising anaerobic parameters.

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Compliance with Ethical Standards

Conflict of interests The authors declare that they have no competing interests.

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